HARDIK SONI

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Education

Indian Institute of Technology, Kharagpur **Dec 2020 – May 2025**
Integrated Bachelors and Masters of Technology in Computer Science *7.74 / 10*

M.P Junior College **Apr 2018 – Apr 2020**
Maharashtra State Board of Secondary and Higher Secondary Education *85.58%*

Awards and Achievements

- Attained a peak rating of **1551 (Specialist)** on **Codeforces** and **1867 (Knight)** on **LeetCode** under the handle **hs094**.
- Secured an **AIR 400** in Joint Entrance Exam(JEE) Advanced 2020 and an **AIR 1952** in Joint Entrance Exam(JEE) Mains 2020.
- Selected as a **Reliance Foundation Scholar** in Artificial Intelligence and Machine Learning, among 10,000+ applicants.
- Collaborated in a team of four to achieve **3rd Rank** in Mathematics Olympiad held at IIT Kharagpur, outperforming 50+ teams.

Experience

Axtria, India | *Software Engineering Internship* | *Smart Data Ingestion Workflow* **May 2024 – July 2024**
Objective: Build a scalable data ingestion pipeline to import, visualize and process large-scale data from remote sources.

- **Conceptualized** a system to detect data source format (**Fixed Width** or **Delimited**) and **implemented** a memory-efficient preview feature for data imported from remote or local sources, allowing users to customize delimiters, column positions, and headers for large data files.
- **Developed** a cloud-sharing service to **streamline** data transfer from **AWS S3**, **Azure Blob Storage**, or **SFTP** via streaming, with additional functionality to unzip and preview ZIP file contents.
- **Diagnosed** and **resolved** critical bugs in DataMAx's Catalog import/export functionality, thereby enhancing overall system reliability and ensuring smooth operations.

Brigham Young University & IIT Kharagpur | *Research Internship* | *UPI Fraud Simulation* **Oct 2022 – July 2023**
Objective: Develop an Swift-based iOS UPI fraud simulator to analyze transaction security, leveraging statistical insights.

- **Developed** a **UPI Simulator** in **Swift (iOS)**, featuring QR code, UPI ID, and contact number functionalities to simulate UPI fraud scenarios, while gathering user insights through targeted interviews.
- Conducted a comprehensive **assessment** of Indian consumer interactions with UPI apps using innovative analytical techniques, uncovering key factors that influence user satisfaction during transactions.
- **Performed** statistical analyses—including the **Mann-Whitney 'U' Test**, **Kruskal-Wallis Test**, and **logistic regression interaction effects**—using Python's **statsmodels** to recommend secure UPI app designs.

Projects

AI-Powered Algorithmic Trading | *LangChain, FastAPI, OpenAI, Groq, MongoDB, Poetry* **Mar 2025 – Present**
Objective: Develop a **multi-agent AI hedge fund** that simulates algorithmic trading using **LLMs** and financial data analysis.

- Designed a **multi-agent system** with AI analysts replicating strategies of top investors like **Warren Buffet** and **Bill Ackman**.
- Integrated **LangChain** with **OpenAI** and **Groq** to generate trading insights and execute back-testing.
- Developed an **asynchronous FastAPI backend** with **MongoDB** for real-time market data processing.
- Implemented a **portfolio manager** and **risk assessment module** to optimize trading decisions and minimize draw-downs.

AI-Powered Blog Post Generator | *FastAPI, MongoDB, LangChain, Groq, OpenAI* **Feb 2025 – Mar 2025**
Objective: Develop an **AI-driven blog generator** that produces **SEO-friendly**, well-structured, markdown-formatted blog posts.

- Implemented a **FastAPI** backend to handle blog post generation and retrieval with **MongoDB** for persistence.
- Integrated **LangChain** with **Groq** and **OpenAI** models to create high-quality, structured content.
- Designed an **asynchronous database handler** using **Motor** to efficiently store and retrieve blog posts.

SmokeCtrl: AI-Driven Healthcare Companion | *QLoRA Finetuning, Flutter, Spring Boot* **Oct 2024 – Feb 2025**
Objective: Develop an **AI healthcare companion** using **RAG** and **QLoRA** for accurate, fair, and low-latency medical responses.

- Developed an **ethical AI** using **RAG** (hybrid retrieval) to reduce misinformation risk by **42%**.
- Fine-tuned LLaMA 3.2 (1.4B) with **LoRA**, achieving **89% accuracy** on tobacco cessation QA with < **500 ms** latency.
- Collaborated with medical experts to ensure fairness, addressing healthcare disparities in low-resource communities.

Relevant Coursework

- Data Structures
- Computer Architecture
- Operating Systems
- Database Management
- Computer Networks
- Systems Programming
- Advanced Image Processing
- Computer Vision
- Probability and Statistics
- Machine Learning
- NLP
- Algorithms - I/II

Technical Skills

Languages: Python, Java, TypeScript, C/C++, SQL, Shell, LaTeX
AI/ML: PyTorch, TensorFlow, Scikit-Learn, LangChain, Sentence-Transformers, OpenAI API, Groq, DeepSeek
Data & Infrastructure: PostgreSQL, MongoDB, Redis, Docker, Kubernetes, AWS (EC2, S3, Lambda), Azure (Functions), Poetry
Frameworks: FastAPI, Spring Boot, Spring MVC, Nest.js, Flask, Express.js, React.js, Next.js
Security & Compliance: OAuth, JWT, XSS Protection, Spring Security